

Homework 6 – Homework

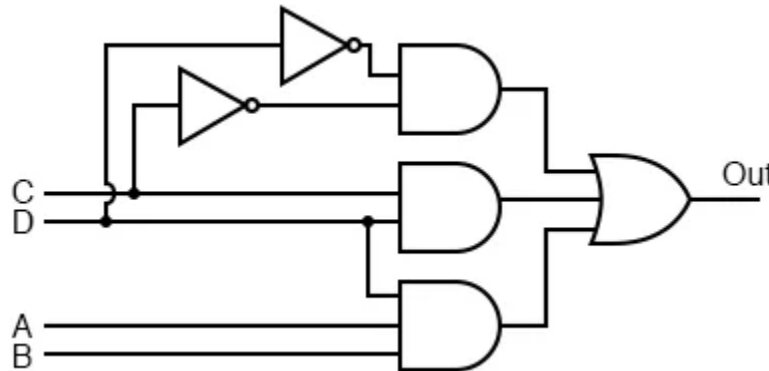
Name:

Section:

Homework Assignment: submit via gradescope

1. Crusty the engineer is a well-intentioned employee in your organization who, unfortunately, is caught in some kind of 70's time warp. Crusty is convinced that a FSM built using discrete logic gates will outperform your shiny new 100Mhz FPGA design. Arguing with Crusty has come to nothing and its time to put an end to this once and for all. You are going to do the calculations to see how fast a discrete logic gate realization can run.

You are to build the FSM using the architecture given in Lecture 9 (see slides 20-26). Use the positive edge triggered flip flops inside the SN74LS273N chip, and the AND, OR and NOT gates inside the SN74LS08N, SN74LS32N, and SN74LS04N chips respectively. You are to assume that the logic inside the MIE and OE boxes are min-term logic such as the example circuit below



To save you time from looking up the chips timing specs from sites like Digi-Key, use these values:

SN74LS273N	D flip flop	setup time = 20nS hold time = 5nS propagation delay = 27nS
SN74LS08N	AND	propagation delay = 27nS
SN74LS32N	OR	propagation delay = 22nS
SN74LS04N	NOT	propagation delay = 22nS

Show your work for partial credit.

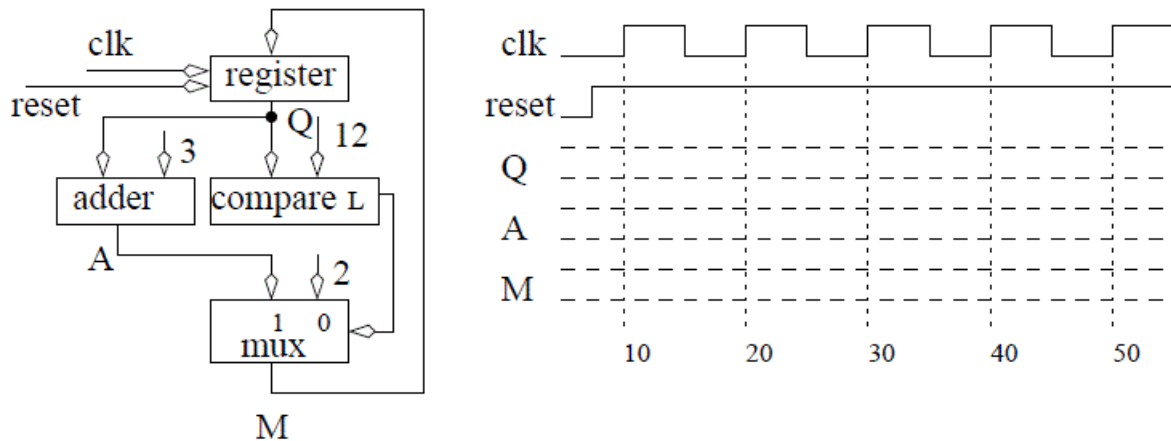
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- e. Assuming you picked the AS series in part d, provide a rough estimate of the clocking frequency of a FSM built from this type of logic. For simplicity assume all gate propagation delays, flipflop setup times, and flipflop propagation delay times are 2 ns. Making sure to show your calculation.

- f. List How does the speed of the FSM in part e compare to using a 100MHz FPGA?

2. Complete the timing diagram below assuming that the register is positive edge trigger and has an asynchronous active low reset.



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Documentation Statement: For all assignments in this course, you may work with any faculty members or students **currently** enrolled in ECE383 unless otherwise indicated. We expect all graded work, to include software programs, wired circuits, lab notebooks, and written reports, to be your own work. If they aren't, you've copied and will receive **no academic credit** even if the copying is documented. Further, copying without attribution is dishonorable and will be dealt with as a suspected honor code violation. As in all courses, cadets must document any assistance received in the execution of graded work. If you receive no assistance on an assignment, the use of the **Documentation: None** statement is mandatory. If no documentation statement exists, the assignment will be returned for correction and the work will be considered at least one day late.

TimeLogs:

How many hours did you spend on this lesson/assignment [include any class time spent on this lesson and assignment, starting from beginning of class on lesson 9 until the beginning of class on lesson 10]

(**Numerical values only**)

3 hours => 3

1 hour and 30 minutes => 1.5

2-3 hours => 2.5

45 minutes => 0.75

If you include anything other than numeric values, you will receive ****no credit**** because we have to manually edit the spreadsheet. Please do not add anything other than numbers.